

BUSINESS INTELLIGENCE AND TOOLS

(24CSH-341)

EXPERIMENT 5: Report Generation using Relational and Multidimensional Data using Microsoft Power BI

AIM To generate relational and multidimensional reports using Microsoft Power BI by utilizing transactional and dimensional data models, enabling effective business analysis, reporting, and decision-making.

TOOLS & SOFTWARE REQUIRED

- Tableau Desktop / Tableau Public
 - Global Superstore Dataset.xlsx
 - Tableau Data Source Page
 - Tableau Prep Builder
 - Tableau Worksheets and Dashboards
 - Tableau Calculated Fields and Filters
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THEORY

Business Intelligence reporting transforms raw data into meaningful information for decision-making. Reports can be broadly classified into Relational Reports and Multidimensional Reports.

Relational Reports

Relational Reports are based on detailed transactional records and generally present information in a tabular format. They are useful for operational analysis and examining individual business transactions.

In Tableau, relational reports can be created using text tables and worksheets containing detailed fields such as Order ID, Customer Name, Product Name, Region, Sales, and Profit.

Multidimensional Reports

Multidimensional Reports are based on OLAP concepts and dimensional models. They allow users to analyze data across multiple dimensions such as Region, Product, Customer, and Time.

In Tableau, multidimensional reports can be created using hierarchies, filters, charts, dashboards, and drill-down operations. These reports help users analyze aggregated business performance from different perspectives.

Tableau supports both reporting approaches through:

- Text tables
 - Cross-tab reports
 - Bar charts
 - Line charts
 - Column charts
 - Filters
 - Parameters
 - Hierarchies
 - Drill-down and drill-up operations
 - Interactive dashboards
 - Calculated fields
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PROCEDURE

Part A: Generate Relational Report

Step 1: Open the Tableau Workbook

Open the Tableau workbook created in the previous experiments.

Step 2: Connect the Dataset

1. Open the Data Source page.
2. Connect the Global Superstore Dataset.xlsx file.
3. Check the data types and field names.
4. Ensure that the required fields are available.

Step 3: Create a Text Table

1. Create a new worksheet.
2. Select the Text Table or Crosstab view.

3. Add the following fields:

- Order ID
- Customer Name
- Product Name
- Region
- Sales
- Profit

Step 4: Apply Sorting and Filtering

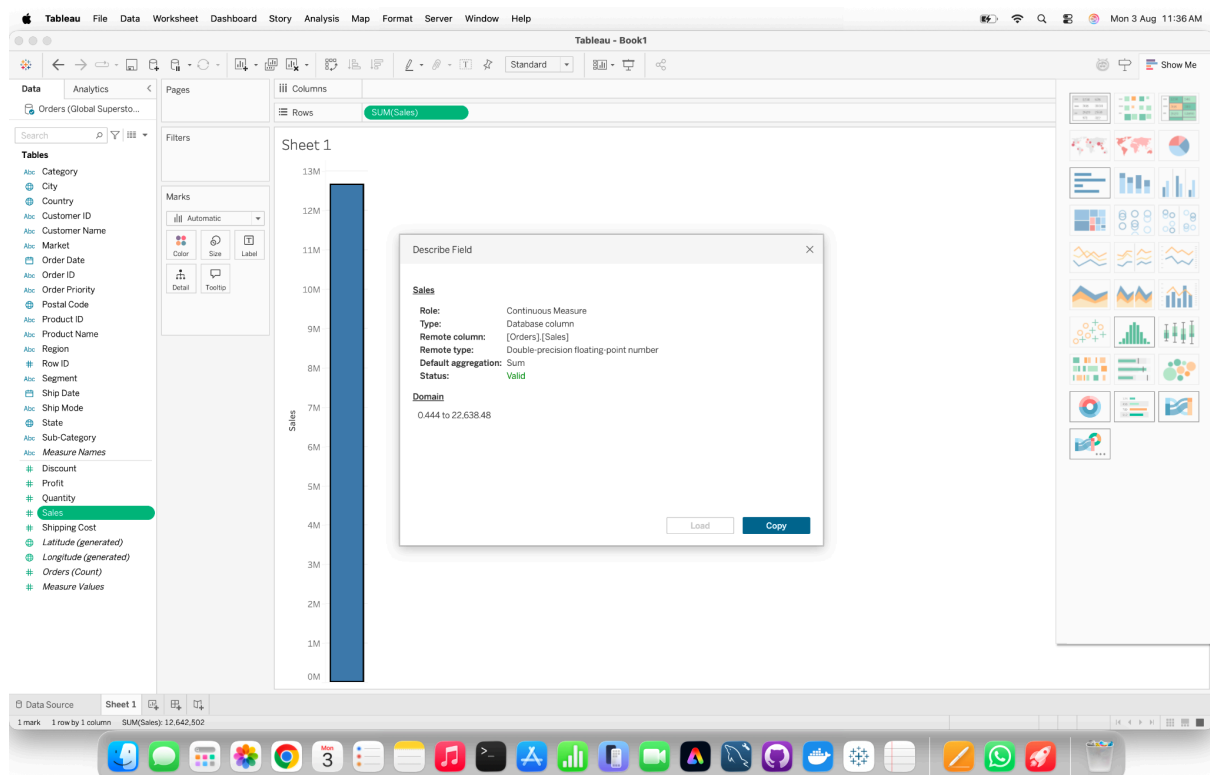
1. Sort the data according to Sales or Profit.
2. Add filters for Region, Category, or Year.
3. Select the required values.
4. Observe the filtered transactional records.

Step 5: Export the Report

1. Open the worksheet.
2. Select Worksheet or Download.
3. Export the report as an Excel file or CSV file.
4. Save the generated report.

Step 6: Record Observations

Observe the detailed transaction information, sales values, profit values, customer details, and regional performance.



Part B: Generate Multidimensional Report

Step 7: Create a Geographic Hierarchy

Create a hierarchy in Tableau using the following fields:

Region → State → City

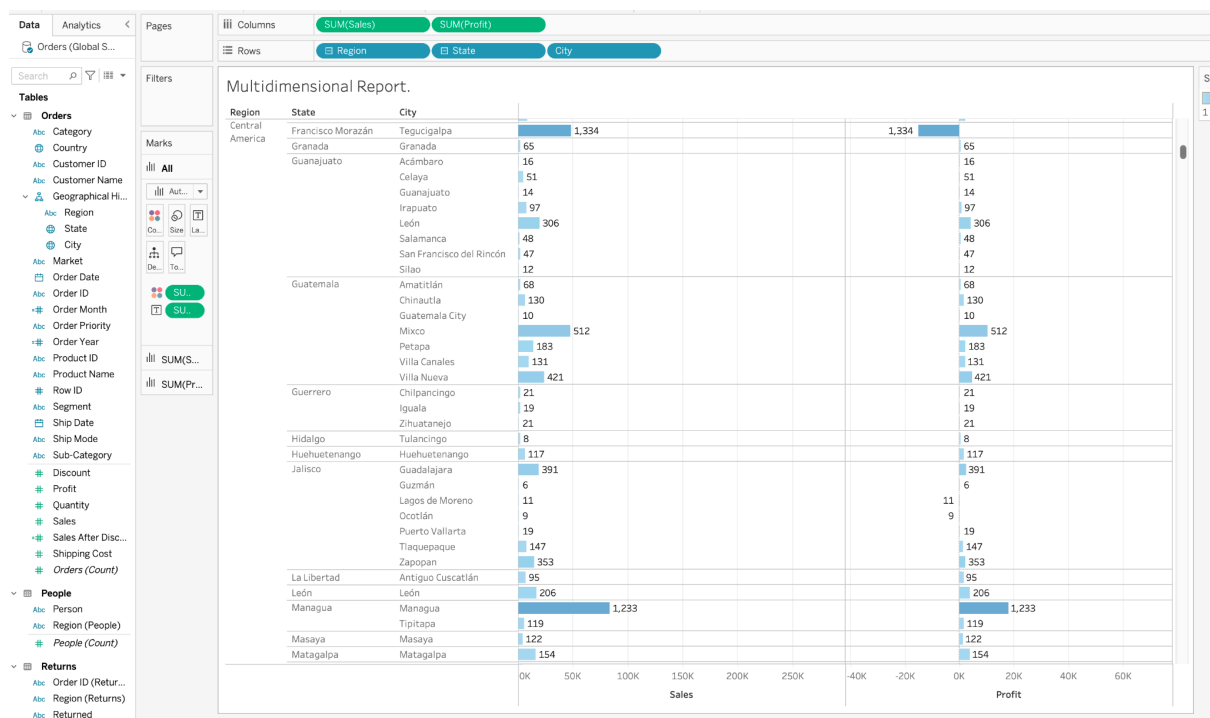
The hierarchy can be created by dragging the fields together in the Data pane and arranging them in the required order.

Step 8: Create a Cross-tab or Matrix-style Report

1. Create a new worksheet.
2. Use a Text Table or Crosstab view.
3. Add the Region → State → City hierarchy to the Rows shelf.
4. Add the following measures to the view:
 - Sales
 - Profit
 - Quantity

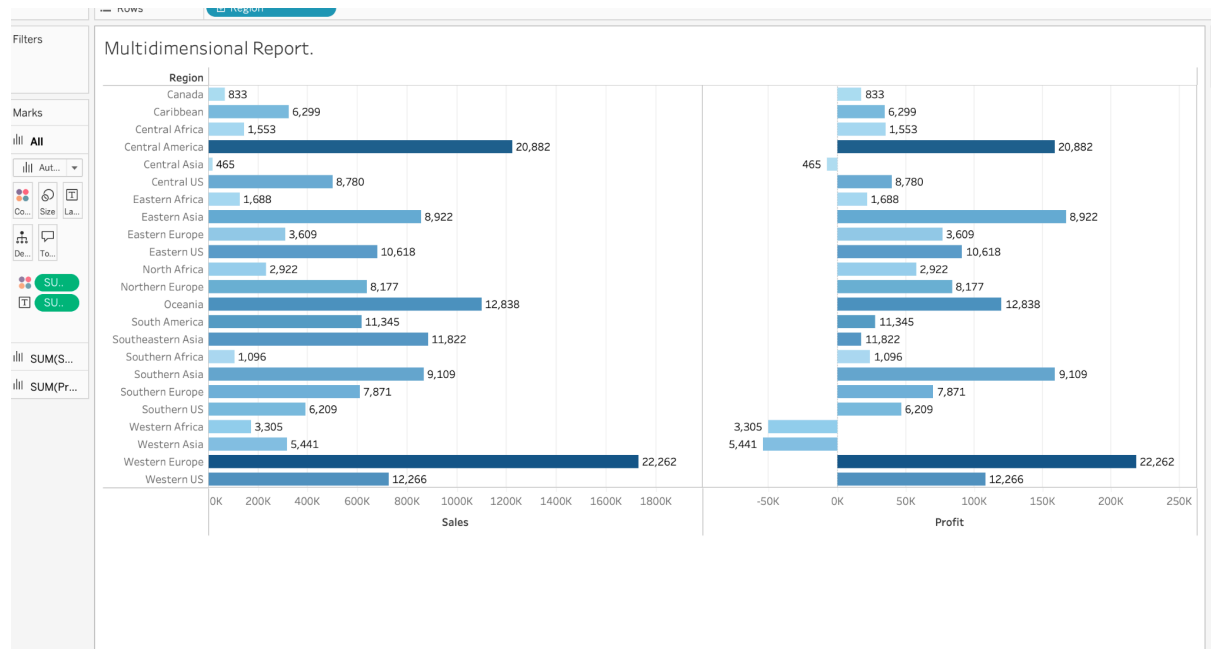
Step 9: Perform Drill-Down

1. Start with the Region level.
2. Click the “+” icon to expand the hierarchy.
3. Move from Region to State.
4. Continue from State to City.
5. Analyze the detailed sales and profit values.



Step 10: Perform Drill-Up

1. Start from the City level.
2. Click the “–” icon to collapse the hierarchy.
3. Move from City to State and then to Region.
4. Observe the summarized business performance.



Step 11: Analyze Aggregated Performance

Analyze the sales, profit, and quantity values across different geographical levels. Identify the regions, states, and cities with higher or lower business performance.

Part C: Comparative Analysis Report

Step 12: Create Required Visualizations

Create separate Tableau worksheets for the following reports:

- Sales by Category
- Profit by Region
- Top 10 Customers by Sales
- Year-wise Sales Trend

Step 13: Use Suitable Chart Types

Use the following visualizations:

- Bar Chart for Sales by Category
- Bar Chart for Profit by Region

- Horizontal Bar Chart for Top 10 Customers by Sales
- Line Chart for Year-wise Sales Trend

Step 14: Apply Filters and Sorting

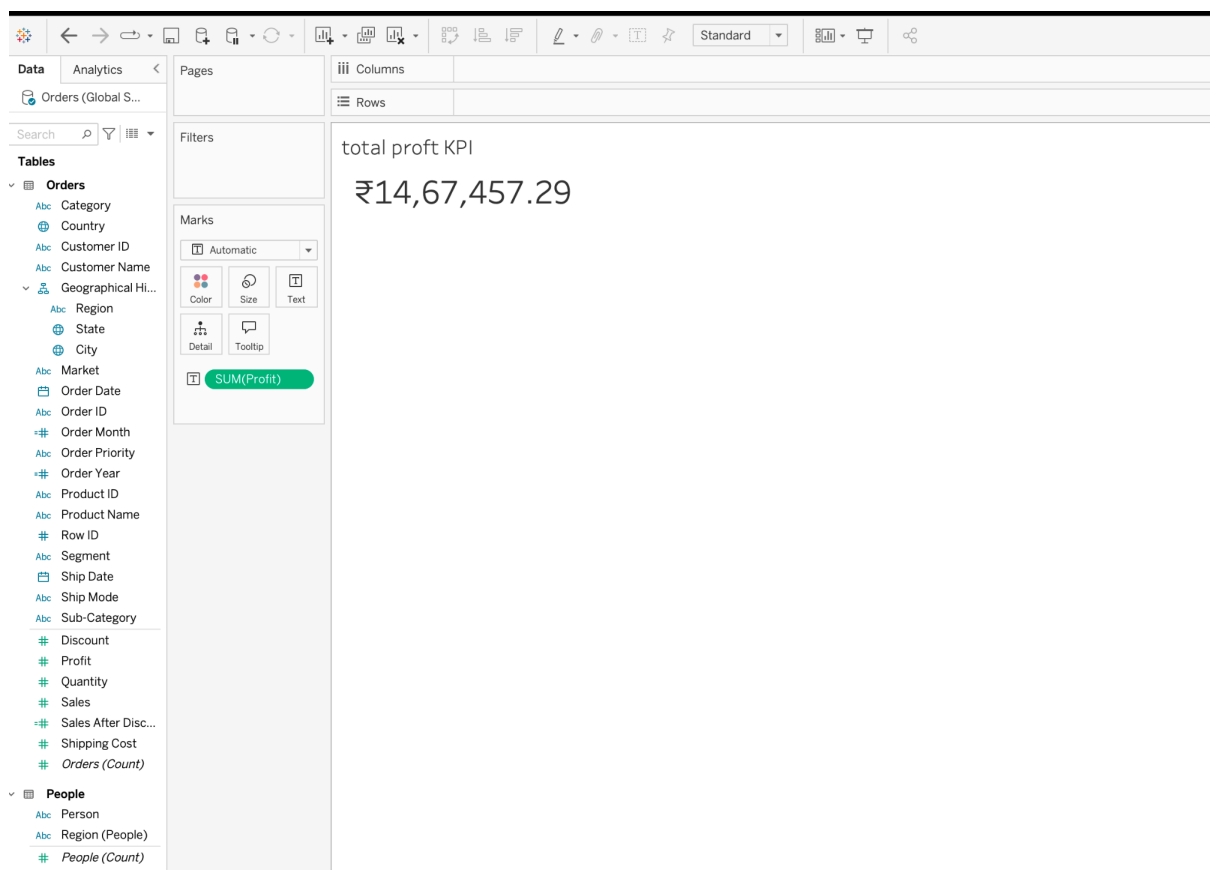
1. Apply filters for Region, Category, and Year.
2. Sort values in descending order where required.
3. Use the Top N filter to display the top 10 customers based on Sales.
4. Add labels and tooltips for better understanding.

Step 15: Compare the Reports

Compare the insights obtained from relational and multidimensional reports.

Relational reports provide detailed transaction-level information, while multidimensional reports provide summarized and aggregated information across different dimensions.

Part D: Create an Interactive Dashboard

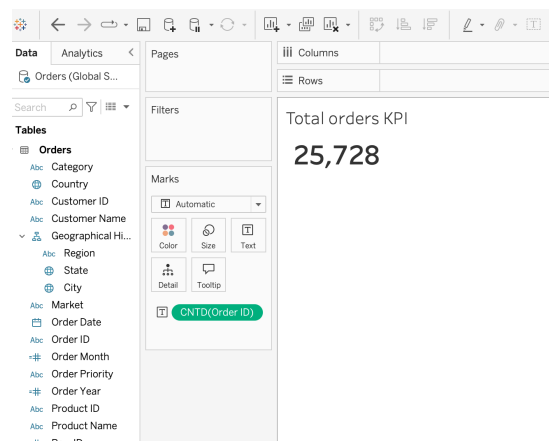
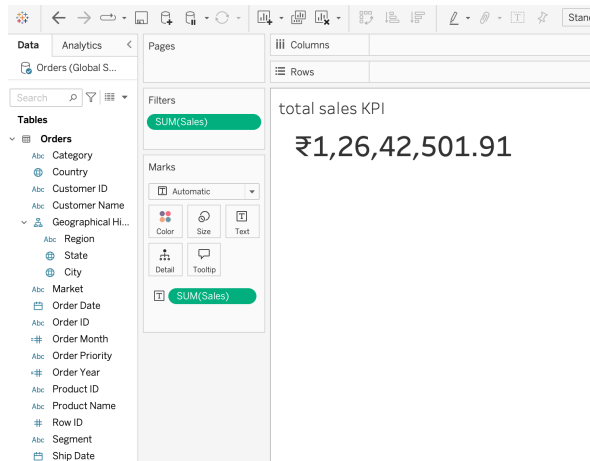


Step 16: Create KPI Reports

Create worksheets or KPI cards for:

- Total Sales
- Total Profit
- Total Orders

Calculated fields may be used where required. For example, the number of orders can be calculated using a distinct count of Order ID.



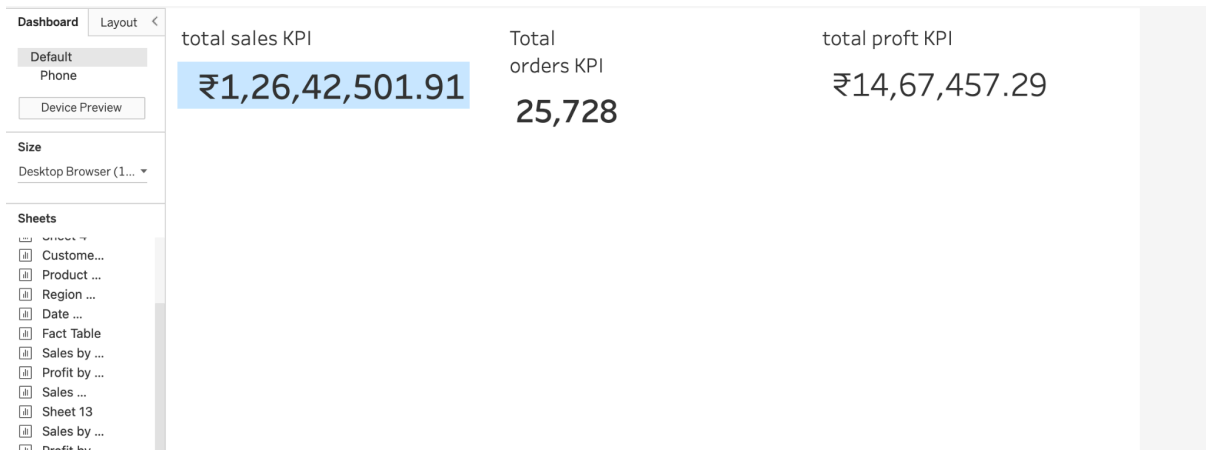
Step 17: Add Filters

Add interactive filters for:

- Region
- Category
- Year

Step 18: Create the Dashboard

1. Open the Dashboard tab.
2. Create a new dashboard.
3. Add the KPI cards.
4. Add the charts created in Part C.
5. Arrange the worksheets in a suitable layout.
6. Add a title and legends.
7. Format the dashboard for better readability.



Step 19: Verify Dashboard Interactions

1. Select different values from the filters.
2. Check whether all visualizations respond correctly.
3. Verify that the KPI cards and charts update according to the selected filters.
4. Use dashboard actions if required to allow interaction between worksheets.

RESULT

OLAP operations were successfully implemented in Tableau using the Global Superstore dataset. Filters and geographical hierarchies enabled interactive analysis of Sales, Profit, Orders, Category, Region, and Year for effective business reporting and decision-making.

LEARNING OUTCOMES

1. Understood the concept of OLAP and its role in Business Intelligence.
2. Learned to perform multidimensional data analysis using Tableau.
3. Learned to implement Slicing using filters.
4. Learned to implement Dicing using multiple dimensional filters.
5. Learned to create and use hierarchies for OLAP analysis.
6. Learned to perform Drill Down from summarized data to detailed data.
7. Learned to perform Drill Up from detailed data to summarized data.
8. Gained practical experience in creating interactive Tableau dashboards for business reporting and decision-making.

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